INCH-POUND

MIL-DTL-15382D
4 August 2021
SUPERSEDING
MIL-B-15382C
25 May 1967

DETAIL SPECIFICATION

BOLT, FIREBRICK ANCHOR

1. SCOPE

1.1 Scope. This specification covers anchor bolts for anchoring firebricks in oil-fired boilers of Naval vessels.

1.2 Classification. Anchor bolts will be of the following classes and types, as specified (see 6.1):

Class A – 25 percent chromium - 20 percent nickel corrosion resistant steel.

Class B – Nickel-chromium-iron-alloy
Type a – Straight end.
Type b – Hook end.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5306

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2.2 Government documents.

2.2.1 Specifications, standards, handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENCE STANDARDS

MIL-STD-1916 – DOD Preferred Methods for Acceptance of Product.

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM A276/A276M – Stainless and Heat-Resisting Steel Bars and Shapes.

ASTM B166 – Standard Specification for Nickel-Chromium-Aluminum Alloy, Nickel-Chromium-Iron Alloys, Nickel-Chromium-Cobalt-Molybdenum Alloy, Nickel- Iron-Chromium-Tungsten Alloy, and Nickel-Chromium- Molybdenum-Copper Alloy Rod, Bar, and Wire.

ASTM E1282 – Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials.

(Copies of these documents are available from www.astm.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331 – Zero Acceptance Number Sampling Plans.

(Copies of this document are available from www.asq.org.)

3. REQUIERMENTS

3.1 Material. Anchor bolts shall be made of material conforming to ASTM A276/A276M, type 310 for class A bolts, and ASTM B166 for class B bolts.

3.2 Type a, straight end. The anchor bolts shall be of the shape and dimensions shown on figure 1. The length of the shank shall be as specified (see 6.1).

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3.3 Type b, hook end. The anchor bolts shall be of the shape and dimension shown on figure 2. The axis of the bent portion of the shank shall be parallel to a diagonal of the head as shown on figure 2. The length of the shank shall be as specified (see 6.1).

3.4 Dimension tolerances. Dimensions of bolts shall be within the tolerance limits specified in table I.

TABLE I. Dimension Tolerances

Item	Dimension	Tolerance
	Inches	Inch
Head width	1.25 square	± 0.0625
Head thickness	0.25	+ 0.06, - 0.01
Shank diameter	0.375	± 0.01
Bolt length	as specified	± 0.0625
Length of hook end, type b only	as specified	± 0.0625

3.5 Heads. Unless otherwise specified (see 6.1), anchor bolt heads shall be forged, resistance (flash) welded, or screwed at the option of the contractor.

3.5.1 Forged. The forged bolts shall have a fillet under the head of not less than 0.125 inches nor more than 0.25 inches in radius.

3.5.2 Resistance (flash) welded. The bolts shall have heads securely flash-welded to the shank, and the weld shall withstand the test specified in 4.5.2.

3.5.3 Screwed. The bolts shall have heads screwed onto the shanks. The heads, through which 0.375-inch 16NC2 or 16UNC2B holes have been tapped, shall be screwed tightly onto shanks, the upper ends of which have been threaded for a distance of approximately 0.4375 inch with 0.375-inch 16NC3 or 16UNC2A threads. The part of the shanks which protrudes through the head shall then be heated and peened over so that the peened portion of the shank overlaps the bolt head such that it is impossible to unscrew the head from the shank (see figure 3).

4. VERIFICATIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Inspection Lot.

4.2.1 Bolts with forged or screwed heads. All bolts of the same type, class, and nominal length, produced under essentially the same condition and offered for inspection at one time shall constitute a lot for purpose of inspection and tests.

4.2.2 Bolts with welded heads. All bolts of the same type, class, and nominal length, fabricated on the same welding machine with the same welding parameters in a single production run and offered for inspection at one time shall constitute a lot for the purposes of inspection and tests.

4.3 Sampling.

4.3.1 Sampling for visual and dimensional examination. Unless otherwise specified in the contract (see 6.1) for sampling for visual and dimensional examination, an "acceptance of zero" (c=0) sampling plan such as MIL-STD-1916 at verification level III or ASQ H1331 at index value of 1.5 shall apply.

4.3.2 Sampling for chemical analysis.

4.3.2.1 When verifiable evidence has been furnished by the manufacturer that the bar and rod stock used conforms to class A or B (see 3.1), as specified, two bolts for chemical analysis shall be taken from each lot to verify their nickel and chromium content.

4.3.2.2 When finished bolts cannot be identified as being processed from bar and rod stock conforming to class A or B (see 3.1) as specified, a sample of bolts shall be selected at random in accordance with table II for chemical analysis.

TABLE II. Sampling for chemical analysis and for tension load test of welded bolts.

Number of bolts in inspection lot	Number of bolts in sample
280 or less	3
281 – 1200	5
1201 – 10,000	8
10,000 or over	13

4.3.2.3 Representative millings or chips shall be taken from the head and from the shank of each sample bolt (from the head or the shank or each bolt in the case of forged bolts) and kept separate. In the case of welded bolts, samples for chemical analysis may be taken from tension load test specimens after tension testing. The samples shall be free from oil, dirt, grit, or other foreign matter and shall consist not less than 2 ounces.

4.3.3 Sampling for tension load test (welded bolts only). A random sample of welded bolts for tension load test shell be selected in accordance with table II.

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4.4 Visual and dimensional examination.

4.4.1 Procedure. Each of the sample bolts selected in accordance with 4.3.1 shall be examined for defects in appearance and dimensions in accordance with table III:

Table III. Possible defects in sample bolts.

Examine	Defects
Appearance	Not type specified. Not shaped as shown on figure 1 or figure 2, as specified. Not clean, free of chips or dirt. Bolt head not fully formed (forged bolts). Head and shank not tapped and threaded as specified (screwed bolts). Protruding part of shank not peened over (screwed bolts)
Dimensions	Dimensions not within tolerances specified (see 3.4)

4.4.2 Acceptance criteria. Any bolt in the sample containing one or more visual or dimensional defects shall be cause for rejection, and if the number of defective bolts in any sample exceeds the acceptance number for the appropriate sampling plan of either MIL-STD-1916 or ASQ H1331, this shall be cause for rejection of the lot represented by the sample. Rejected lots may again be offered for inspection provided the contractor has reexamined each item of the lot and removed or reworked nonconforming bolts, and states that he has examined the entire lot and found it to conform with the requirements of this specification.

4.5 Tests.

4.5.1 Chemical analysis.

4.5.1.1 Procedure (applicable to 4.3.2.1 samples). The samples obtained as specified in 4.3.2.1 shall be tested for nickel and chromium content in accordance with ASTM E1282.

4.5.1.1.1 Acceptance criteria. If any sample fails to conform to the nickel or chromium content specified in 3.1 for class A or B bolts, as applicable, this shall cause for rejection of the lot from which the samples were taken.

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4.5.1.2 Procedure (applicable to 4.3.2.2 samples). All samples selected in accordance with 4.3.2.2 shall be tested for content of nickel and chromium in accordance ASTM E1282. Two of the samples selected in accordance with 4.3.2.2 shall be tested for specified elements other than nickel and chromium in accordance with ASTM E1282.

4.5.1.2.1 Acceptance criteria. If any analysis result indicates nonconformance with a requirement of 3.1 for the class specified, this shall be cause for rejection of the lot from which the samples were taken.

4.5.2 Tension load test (welded bolts). Each of the sample welded bolts selected in accordance with 4.3.3 shall be subjected to a tension load of 5000 pounds. The load shall be applied in such a manner as to subject the welded joint to the tension load.

4.5.2.1 Acceptance criteria. If any bolt fails to withstand the tension load, this shall be cause for rejection of the lot represented by the sample.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Length of shank required, and if type b, length of the hook end (see 3.2 and 3.3).
(d) Weather head of bolt should be other than specified (see 3.5).
(e) Verification Level or Index Value (see 4.3.1)
(f) Packaging (see 5.1)

6.2 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this

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document based on the entire content irrespective of the marginal notations and relationship to the previous issue."

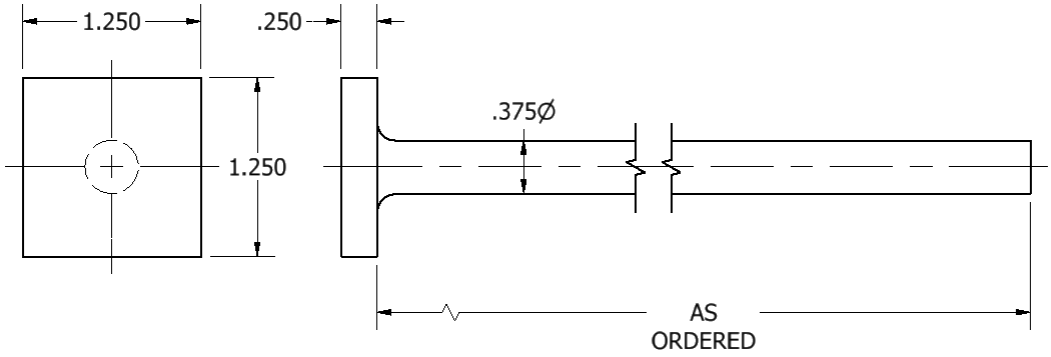


FIGURE 1. Type a, straight end.

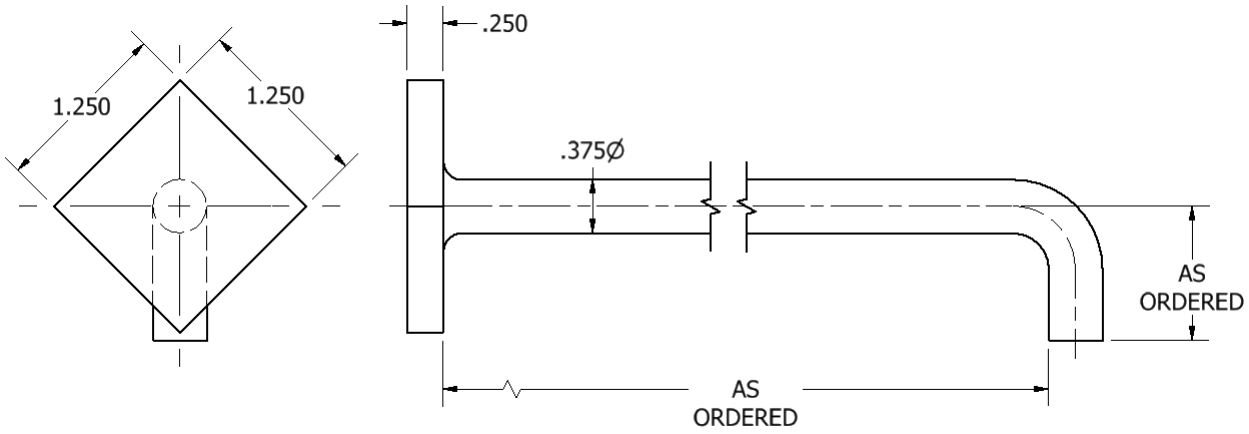


FIGURE 2. Type b, hook end.

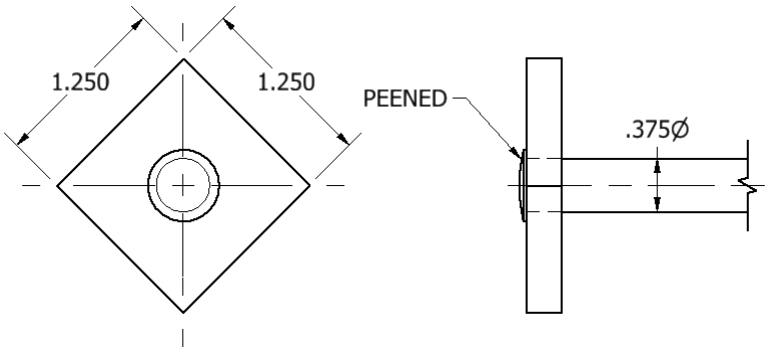


FIGURE 3. Screwed and peened heads for type a or b anchor bolts.

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CONCLUDING MATERIAL:

Custodians:
Navy - SH
DLA - GS

Preparing Activity:
DLA – GS7

(Project 5306-2021-022)

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